MAP2341

Notes on FEM for Poisson's Eqn. with Neumann BCs in 1D.

$$-u'' = f \quad \text{on } (0,1)$$

$$u'(0) = a$$

$$u'(1) = b$$

Scratch work:

$$-u''v = fv$$

IBP on $(0,1)$

$$(u'v)' = u''v + u'v'$$

$$\Rightarrow u''v = -u'v' + (u'v)'$$

$$\int_0^1 u'v' - [u'v]_{x=0}^{x=1} = \int_0^1 fv$$

We do not want to kill this term. So we don't want the test functions to vanish on the boundary.

Weak formulation:

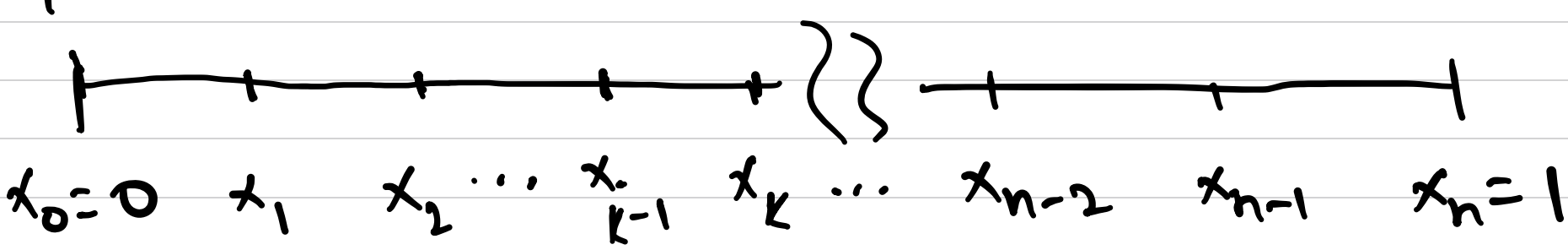
Find $u \in H^1(0,1)$ s.t.

$$\int_0^1 u' v' - \left[b\, v(1) - a\, v(0) \right] = \int_0^1 f v$$

$$\forall\ v \in H^1(0,1)$$

Finite Element formulation:

Let



$I_k := [x_{k-1}, x_k]$; $h_k := x_k - x_{k-1}$;

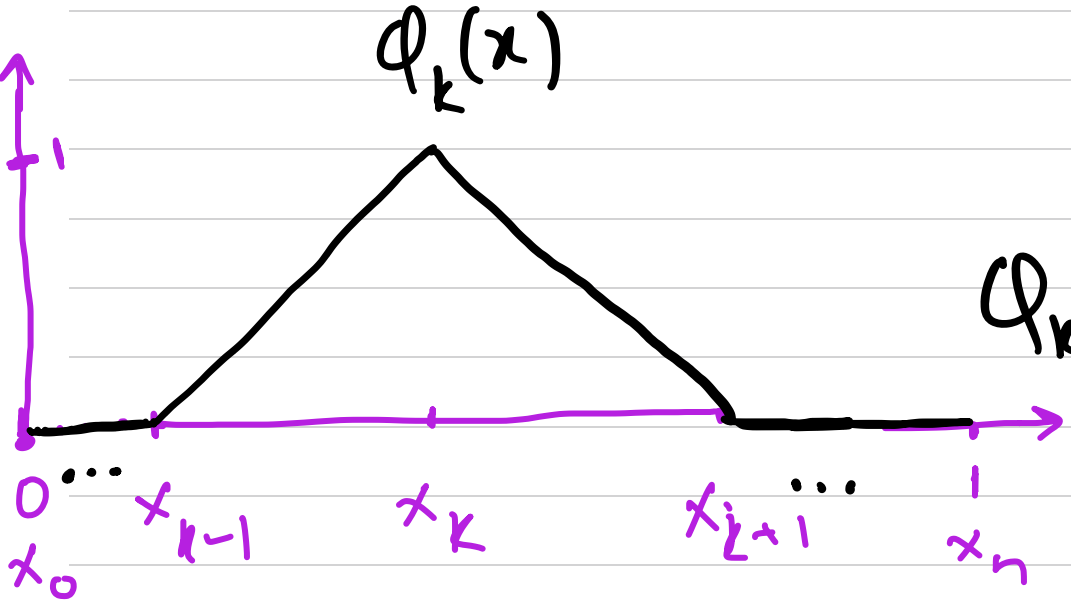
$$h := \max_{1 \le k \le n} \{ h_k \}$$

$$V_h := \left\{ v \in L^2(0,1) \cap C(0,1) : v\big|_{I_k} \in \mathbb{P}_1(I_k) \quad \forall\ 1 \le k \le n \right\}$$

$\downarrow$

Finite Element Space

Define the basis for V_h s.t.



$$\varphi_k(x) = \begin{cases} \dfrac{x - x_{k-1}}{h_k}; & x \in I_k \\[2ex] \dfrac{x_{k+1} - x}{h_{k+1}}; & x \in I_{k+1} \\[2ex] 0 & \text{everywhere else} \end{cases}$$

NOTE: For a uniform mesh

$$h_k = h = \frac{1-0}{n} \quad \forall\ 1 \le k \le n$$

Find $u_h \in V_h$ s.t

$$\sum_{k=1}^{n} \int_{x_{k-1}}^{x_k} u_h' v_h' - \left[b\, v_h(1) - a\, v_h(0) \right] = \sum_{k=1}^{n} \int_{x_{k-1}}^{x_k} f v_h \quad \forall\ v_h \in V_h$$

Finite Element formulation in Matrix-Vector form:

Scratch Work:

Since $u_h, v_h \in V_h$ say

$$u_h = \sum_{j=0}^{n} u_j \varphi_j \quad \text{and} \quad v_h = \sum_{i=0}^{n} v_i \varphi_i$$

Then we have

$$\sum_{k=1}^{n} \int_{x_{k-1}}^{x_k} \left(\sum_{j=0}^{n} u_j \varphi_j\right)' \left(\sum_{i=0}^{n} v_i \varphi_i\right)' dx$$

$$- b \sum_{i=0}^{n} v_i \varphi_i(1) + a \sum_{i=0}^{n} v_i \varphi_i(0)$$

$$= \sum_{k=1}^{n} \int_{x_{k-1}}^{x_k} f \sum_{i=0}^{n} v_i \varphi_i \, dx$$

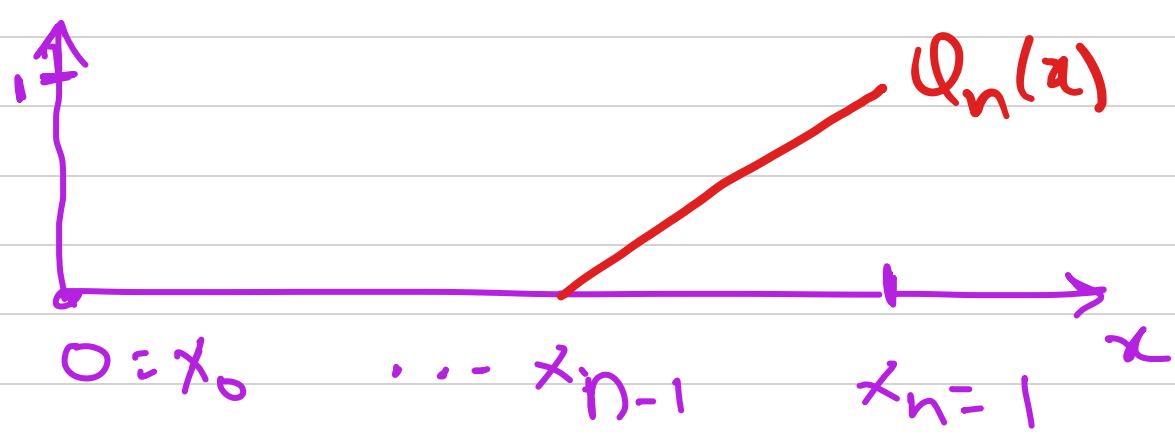
Therefore

$$\sum_{k=1}^{n} \int_{x_{k-1}}^{x_k} \left(\sum_{j=0}^{n} u_j \varphi_j\right)' \left(\sum_{i=0}^{n} v_i \varphi_i\right)' dx$$

$$= \sum_{k=1}^{n} \int_{x_{k-1}}^{x_k} f \sum_{i=0}^{n} v_i \varphi_i \, dx + b \sum_{i=0}^{n} v_i \varphi_i(1)$$

$$- a \sum_{i=0}^{n} v_i \varphi_i(0)$$

Notice that $\sum_{i=0}^{n} v_i \varphi_i(1) = v_n$



Likewise $\sum_{i=0}^{n} v_i \varphi_i(0) = v_0$

Then $$\sum_{k=1}^{n} \int_{x_{k-1}}^{x_k} \left(\sum_{j=0}^{n} u_j \varphi_j\right)' \left(\sum_{i=0}^{n} v_i \varphi_i\right)'$$

$$= \sum_{k=1}^{n} \int_{x_{k-1}}^{x_k} f \sum_{i=0}^{n} v_i \varphi_i \, dx + b v_n - a v_0$$

Finite element formulation in the matrix-vector form:

find $u \in \mathbb{R}^{n+1}$ such that

$$v^T A u = v^T f \qquad \forall v \in \mathbb{R}^{n+1}$$

where

$$A_{ij} = \sum_{k=1}^{n} \int_{x_{k-1}}^{x_k} \varphi_j' \varphi_i' \, dx \qquad 0 \le i, j \le n$$

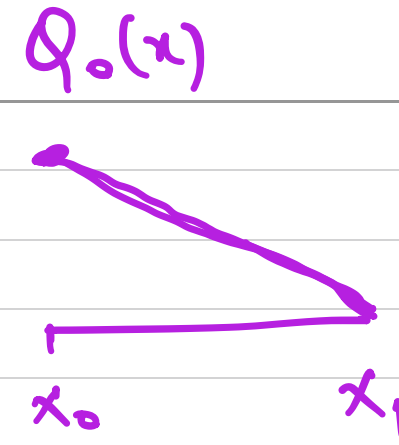
$$f_i = \sum_{k=1}^{n} \int_{x_{k-1}}^{x_k} f \varphi_i \, dx \qquad 1 \le i \le n-1$$

$$f_0 = \sum_{k=1}^{n} \int_{x_{k-1}}^{x_k} f \varphi_0 \, dx - a \; ; \; f_n = \sum_{k=1}^{n} \int_{x_{k-1}}^{x_k} f \varphi_n \, dx + b$$

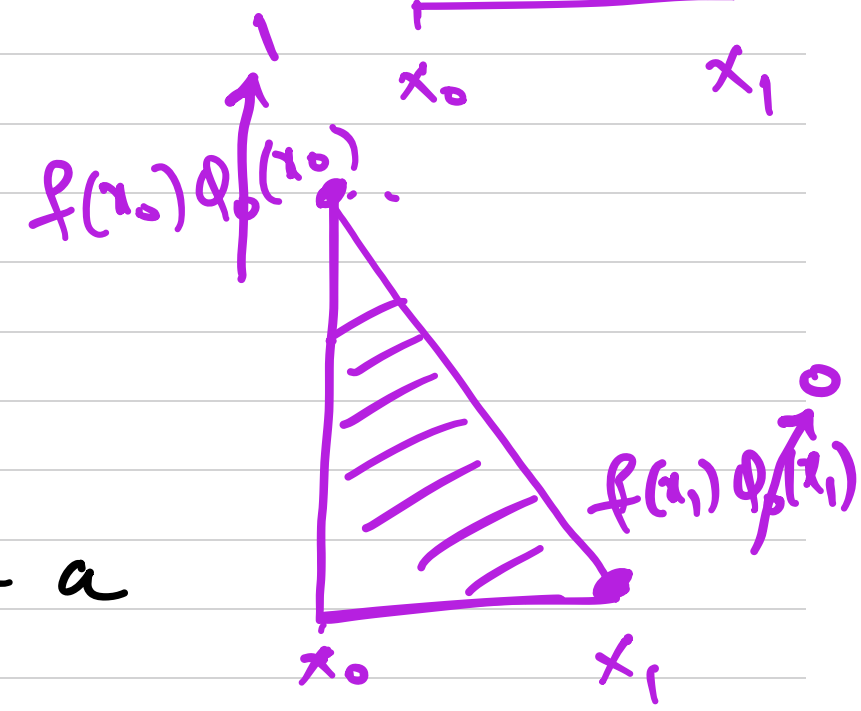
That is Solve for $u \in \mathbb{R}^{n+1}$ where

$$\begin{bmatrix} 0 & \cdots & 0 & \frac{x_0}{\Theta_0} - \frac{x_1}{\Theta_1} & \frac{x_0}{\Theta_0} - \frac{x_1}{\Theta_1} \\ \vdots & & & \frac{x_0}{\Theta_0} - \frac{x_1}{\Theta_1} & \frac{x_0}{\Theta_0} - \frac{x_1}{\Theta_1} \\ \vdots & & & \frac{x_1}{\Theta_1} - \frac{x_2}{\Theta_2} & \frac{x_0}{\Theta_0} - \frac{x_1}{\Theta_1} + \frac{x_2}{\Theta_2} \\ \vdots & & & \frac{x_1}{\Theta_1} - \frac{x_2}{\Theta_2} & \frac{x_1}{\Theta_1} - \frac{x_2}{\Theta_2} \\ \vdots & & & \frac{x_1}{\Theta_1} - \frac{x_2}{\Theta_2} + \frac{x_3}{\Theta_3} & \vdots \\ \vdots & & & \vdots & \vdots \\ \frac{x_{n-1}}{\Theta_{n-1}} - \frac{x_n}{\Theta_n} & \cdots & \cdots & \cdots & 0 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} \frac{x_0}{\Theta_0} - \frac{x_1}{\Theta_1} \\ \frac{x_0}{\Theta_0} - \frac{x_1}{\Theta_1} + \frac{x_2}{\Theta_2} \\ \frac{x_1}{\Theta_1} - \frac{x_2}{\Theta_2} + \frac{x_3}{\Theta_3} \\ \vdots \\ \frac{x_{n-1}}{\Theta_{n-1}} - \frac{x_n}{\Theta_n} + \frac{x_{n+1}}{\Theta_{n+1}} \end{bmatrix}$$

$$f_0 = \sum_{k=1}^{n} \int_{x_{k-1}}^{x_k} f\,\varphi_0\,dx - a$$



$$= \int_{x_0}^{x_1} f\,\varphi_0 - a$$



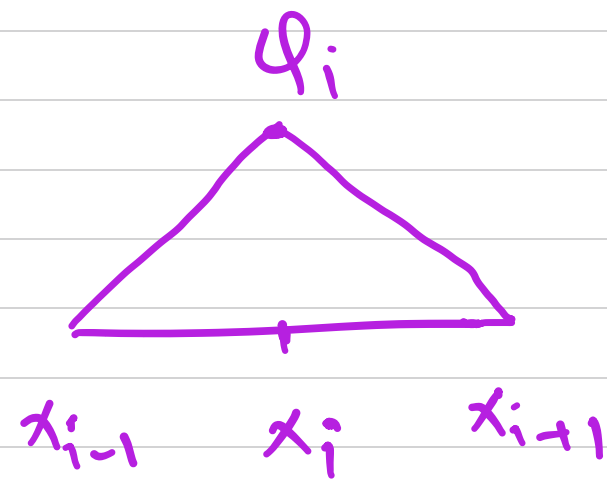
$$\simeq f(x_0)\,\cancelto{1}{\varphi_0(x_0)} * h - a$$

approximating integral by Trapezoidal rule

$$= \frac{1}{2}(x_0 - x_1)\left(f(x_0)\,\cancelto{1}{\varphi_0(x_0)} + f(x_1)\,\cancelto{0}{\varphi_0(x_1)}\right)$$

$$= \frac{1}{2} h\, f(x_0)$$

$$f_i = \sum_{k=1}^{n} \int_{x}^{x_k} f\,\varphi_i\,dx \approx f(x_i) * h \quad 1 \le i \le n-1$$



$$= \int_{x_{i-1}}^{x_i} f\varphi_i + \int_{x_i}^{x_{i+1}} f\varphi_i$$

$$\approx \frac{(f\varphi_i)(x_i) + (f\varphi_i)(x_{i-1})}{2} * (x_i - x_{i-1})$$

approximation by trapezoidal rule

$$+ \frac{(f\varphi_i)(x_{i+1}) + (f\varphi_i)(x_i)}{2} * (x_{i+1} - x_i)$$

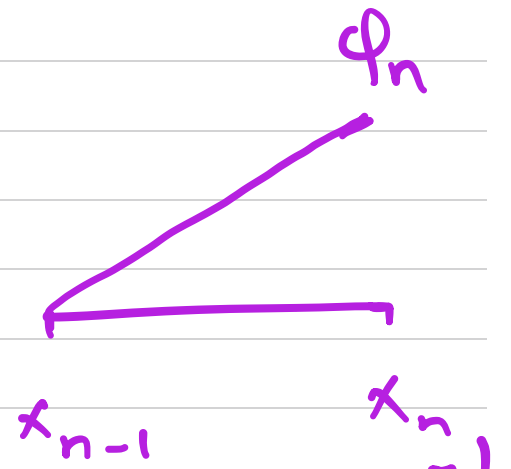
$$= \frac{f(x_i)\,\cancelto{1}{\varphi_i(x_i)} + f(x_{i-1})\,\cancelto{0}{\varphi_i(x_{i-1})}}{2} * h$$

$$+ \frac{f(x_{i+1})\,\cancelto{0}{\varphi_i(x_{i+1})} + f(x_i)\,\cancelto{1}{\varphi_i(x_i)}}{2} * h$$

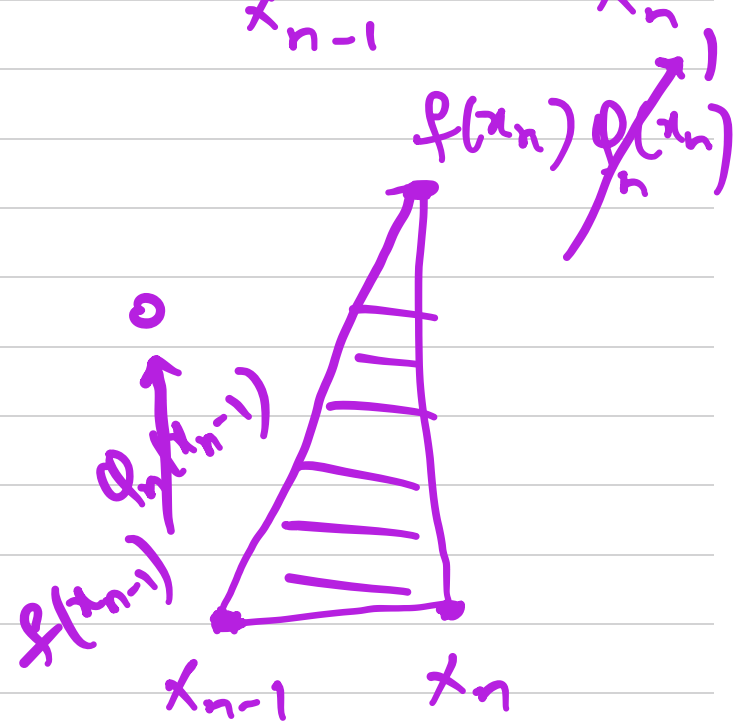
$$= f(x_i) * \frac{h}{2} + f(x_i) * \frac{h}{2}$$

$$= f(x_i) * h$$

$$f_n = \sum_{k=1}^{n} \int_{x_{k-1}}^{x_k} f \varphi_n \, dx + b$$



$$= \int_{x_{n-1}}^{x_n} f \varphi_n \, dx + b$$



$$\simeq f(x_n)\,\cancel{\varphi_n(x_n)}^{\,1} * h + b$$

approximation by trapezoidal rule

$$= \frac{1}{2}(x_n - x_{n-1})\, f(x_n) + b$$

$$= \frac{1}{2} h\, f(x_n) + b$$